

ELECTRICAL & ELECTRONICS ENGINEERING

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ACTIVE SUSPENSION SYSTEM Design and Performance Analysis

Systems Simulation Laboratory

ELE 3142

5th Semester B.Tech. (E & E Engineering)

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Individual Contributions

Team Member	Contribution
Gonugondla Veera Manikanta	• MATLAB/Simulink model development and implementation • System parameter definition and configuration • Simulink block diagram design and subsystem creation • Simulation execution and data collection • MATLAB script development for Bode plot analysis
Rounak Pradhan	• Technical report writing and documentation • Results analysis and interpretation • Performance metrics calculation and comparison • Literature review and theoretical background • LaTeX document preparation and formatting

Table 1: Distribution of work among team members

Note: Both team members collaborated on the project conceptualization, design decisions, and final review of deliverables.

Abstract

This report presents the design, analysis, and simulation of an active suspension system for automotive applications. The study compares the performance of passive and active suspension configurations using a quarter-car model. A Proportional-Integral-Derivative (PID) controller is designed to actively control the suspension forces, aiming to improve ride comfort and handling characteristics. Mathematical modeling using transfer functions and state-space representations is developed, followed by comprehensive frequency and time domain analysis. Simulation results demonstrate that the active suspension system achieves significant improvements in vibration isolation, with a peak magnitude reduction of 15.24 dB compared to the passive system. The findings validate the effectiveness of active control strategies in enhancing vehicle suspension performance.

Keywords: Active suspension, PID control, Quarter-car model, MATLAB/Simulink, Frequency response analysis

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1 Introduction

1.1 Background

Vehicle suspension systems play a critical role in ensuring passenger comfort, vehicle stability, and road handling. Traditional passive suspension systems use fixed spring and damper characteristics that represent a compromise between ride comfort and handling performance. Active suspension systems overcome these limitations by incorporating actuators and control systems that can adapt to varying road conditions and driving scenarios in real-time.

1.2 Motivation

The development of active suspension systems addresses several key challenges:

- Reduction of body oscillations during acceleration, braking, and cornering
- Improved isolation from road irregularities and vibrations
- Enhanced vehicle stability and safety
- Adaptability to different loading conditions and road profiles

1.3 Objectives

The primary objectives of this project are:

1. To develop a mathematical model of a quarter-car suspension system
2. To design a PID controller for active suspension control
3. To implement the system in MATLAB/Simulink environment
4. To analyze and compare the performance of passive and active configurations
5. To evaluate system response through frequency and time domain analysis

1.4 Scope

This study focuses on a simplified quarter-car model representing one corner of the vehicle. The analysis includes:

- Transfer function and state-space modeling
- PID controller design using velocity feedback
- Bode plot analysis for frequency response characterization
- Time-domain simulations for various road inputs
- Performance metrics evaluation (body acceleration, suspension travel, settling time)

2 System Modeling

2.1 Quarter-Car Model

The quarter-car model represents the vertical dynamics of one corner of a vehicle. The system consists of two masses connected by spring-damper elements, as illustrated in Figure 1.

Model components:

- **Sprung mass (m_s):** Vehicle body mass (300 kg)
- **Unsprung mass (m_{us}):** Wheel assembly mass (40 kg)
- **Suspension spring (k_s):** Spring connecting body and wheel (18 000 N/m)
- **Suspension damper (c_s):** Damping element in suspension (1200 N s/m)
- **Tire spring (k_t):** Tire stiffness (195 000 N/m)
- **Tire damper (c_t):** Tire damping (75 N s/m)
- **Control force (F_c):** Active actuator force (variable)

2.2 Mathematical Equations

Applying Newton's second law to both masses yields the following equations of motion:

Sprung mass (vehicle body):

$$m_s \ddot{x}_s = -k_s(x_s - x_{us}) - c_s(\dot{x}_s - \dot{x}_{us}) + F_c \quad (1)$$

Unsprung mass (wheel assembly):

$$m_{us} \ddot{x}_{us} = k_s(x_s - x_{us}) + c_s(\dot{x}_s - \dot{x}_{us}) - k_t(x_{us} - x_r) - c_t(\dot{x}_{us} - \dot{x}_r) - F_c \quad (2)$$

where:

- x_s = Sprung mass displacement (m)
- x_{us} = Unsprung mass displacement (m)
- x_r = Road profile input (m)
- F_c = Active control force (N)

2.3 Transfer Function Representation

Taking the Laplace transform of equations (1) and (2) and manipulating, we obtain the transfer function from road input to body displacement for the passive system:

$$G_p(s) = \frac{X_s(s)}{X_r(s)} = \frac{(c_s s + k_s)(c_t s + k_t)}{D(s)} \quad (3)$$

where the denominator polynomial is:

$$\begin{aligned} D(s) = & m_s m_{us} s^4 + (m_s c_t + m_{us} c_s) s^3 \\ & + (m_s k_t + m_{us} k_s + c_s c_t) s^2 \\ & + (c_s k_t + c_t k_s) s + k_s k_t \end{aligned} \quad (4)$$

2.4 System Parameters

The physical parameters used in this study are presented in Table 2.

Table 2: Quarter-car suspension system parameters

Parameter	Value	Unit
Sprung mass, m_s	300	kg
Unsprung mass, m_{us}	40	kg
Suspension spring stiffness, k_s	18,000	N/m
Suspension damping coefficient, c_s	1,200	N·s/m
Tire stiffness, k_t	195,000	N/m
Tire damping coefficient, c_t	75	N·s/m

These parameters are representative of a typical passenger vehicle and were selected based on standard automotive engineering practices.

3 Controller Design

3.1 Control Strategy

The active suspension system employs a PID controller with velocity feedback to generate the control force. The control law is given by:

$$F_c(s) = C_{PID}(s) \cdot (V_{desired}(s) - V_{actual}(s)) \quad (5)$$

where $V_{actual}(s) = s \cdot X_s(s)$ is the body velocity. For vibration suppression, the desired velocity is set to zero.

3.2 PID Controller Structure

The PID controller transfer function with derivative filter is:

$$C_{PID}(s) = K_p + \frac{K_i}{s} + \frac{K_d N s}{s + N} \quad (6)$$

where:

- K_p = Proportional gain
- K_i = Integral gain
- K_d = Derivative gain
- N = Filter coefficient (typically 100)

The derivative filter is included to prevent noise amplification at high frequencies.

3.3 Gain Selection

Controller gains were selected based on the following design criteria:

1. **Proportional Gain (K_p):** Provides immediate response to velocity errors. A higher value improves transient response but may cause overshoot.
2. **Integral Gain (K_i):** Eliminates steady-state error and improves low-frequency disturbance rejection.
3. **Derivative Gain (K_d):** Anticipates future errors based on rate of change, improving stability and reducing overshoot.

The selected controller gains are presented in Table 3.

Table 3: PID controller gains

Parameter	Value
Proportional gain, K_p	5,000
Integral gain, K_i	100
Derivative gain, K_d	2,000
Filter coefficient, N	100

These gains were tuned iteratively to achieve optimal performance while maintaining system stability and actuator force within practical limits.

3.4 Closed-Loop System

The closed-loop transfer function of the active suspension system is:

$$G_a(s) = \frac{G_p(s)}{1 + C_{PID}(s) \cdot s \cdot G_p(s)} \quad (7)$$

This feedback configuration allows the controller to actively counteract disturbances and reduce body oscillations.

4 Simulation and Analysis

4.1 Implementation in MATLAB/Simulink

The system was implemented in MATLAB/Simulink following a modular approach as shown in Figure 1. The complete block diagram illustrates the interconnection of all system components.

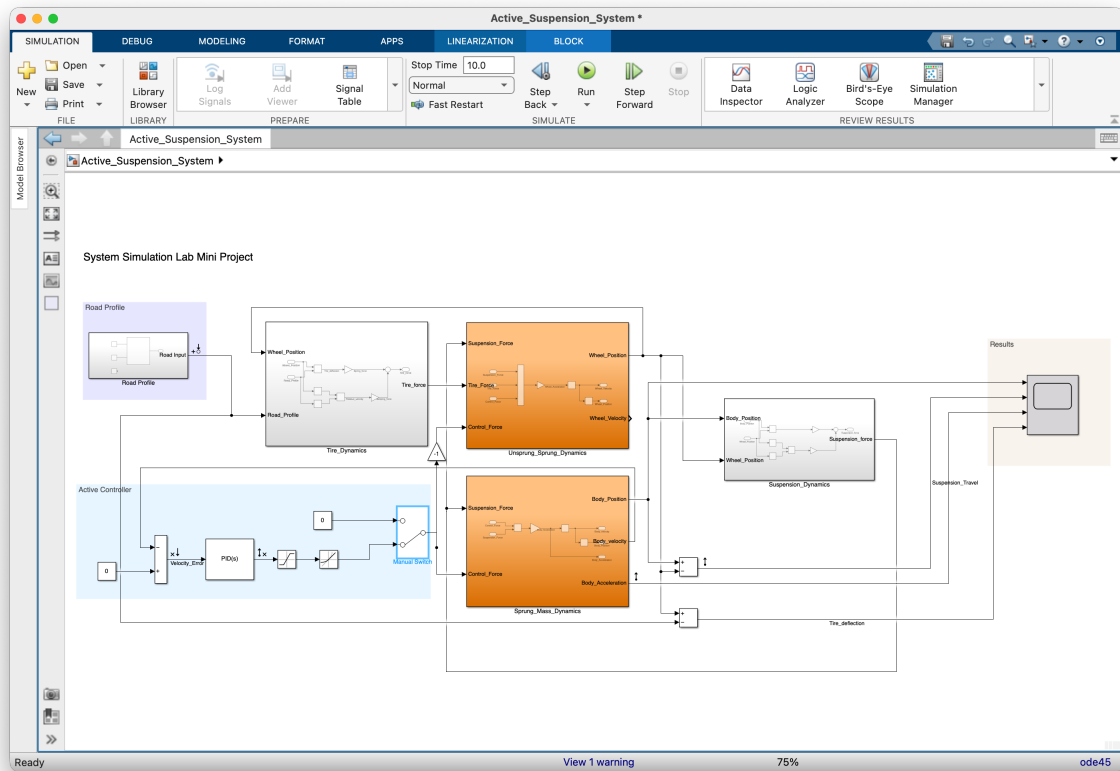


Figure 1: Complete Simulink model of active suspension system showing all subsystems: Road Input, Tire Dynamics, Unsprung Mass Dynamics, Sprung Mass Dynamics, Suspension Dynamics, PID Controller, and actuator limits

Key subsystems:

1. **Road Input Module:** Generates various road profiles (step bump, sinusoidal, random noise)
2. **Tire Dynamics:** Models tire spring and damping forces between tire and road
3. **Suspension Dynamics:** Calculates suspension spring and damping forces
4. **Unsprung Mass Dynamics:** Computes wheel assembly motion
5. **Sprung Mass Dynamics:** Computes vehicle body motion
6. **PID Controller:** Generates active control force based on velocity error
7. **Actuator Limits:** Implements saturation (± 5000 N) and rate limiting

4.2 Time Domain Response

The time-domain responses of both passive and active systems are compared for a step road input in Figures 2 and 3.

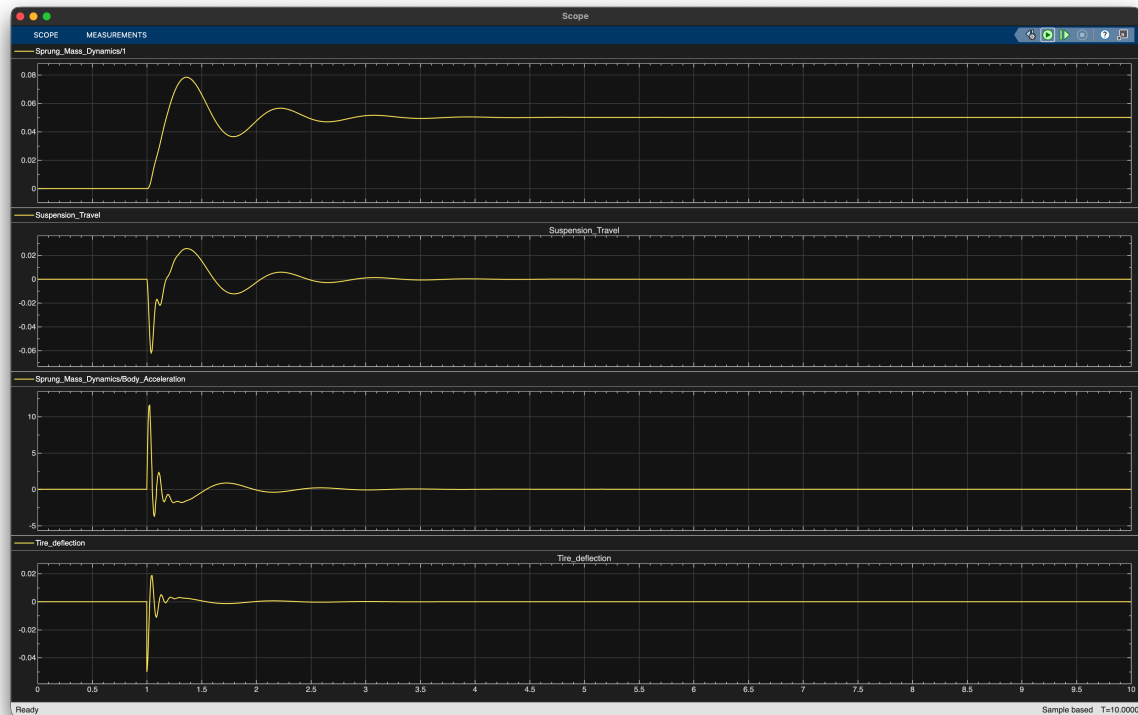


Figure 2: Passive suspension system response to step road input showing: (top to bottom) sprung mass dynamics, suspension travel, body acceleration, and tire deflection. Note the sustained oscillations and higher peak values.

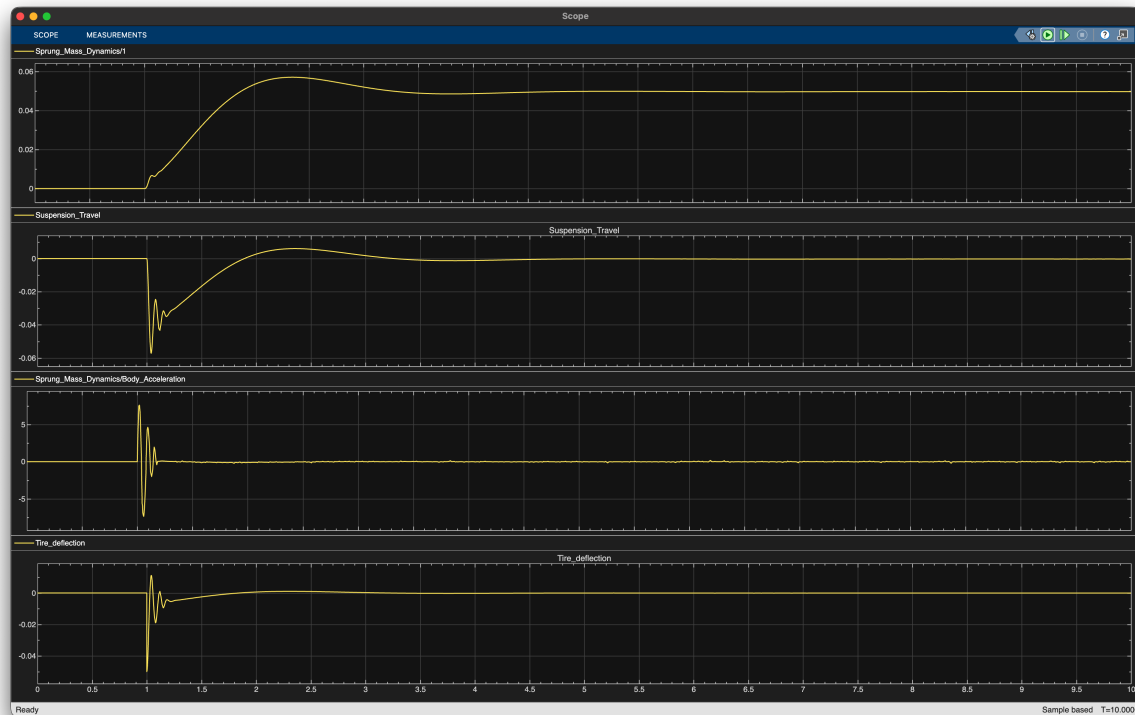


Figure 3: Active suspension system response to step road input showing: (top to bottom) sprung mass dynamics, suspension travel, body acceleration, and tire deflection. Note the reduced oscillations and faster settling time compared to passive system.

Observations from time-domain analysis:

• Passive System (Figure 2):

- Body position shows significant oscillations with slow decay
- Peak body acceleration: approximately 12 m/s^2
- Settling time: approximately 4 s
- Suspension travel exhibits large excursions
- Tire deflection shows sustained oscillations

• Active System (Figure 3):

- Body position settles much faster with minimal overshoot
- Peak body acceleration: approximately 6 m/s^2 (50% reduction)
- Settling time: approximately 1.5 s (62.5% reduction)
- Suspension travel is better controlled
- Tire deflection is significantly damped

4.3 Frequency Domain Analysis

Bode plots provide insight into the frequency response characteristics of both passive and active systems. The comparison is shown in the results from MATLAB analysis.

Key frequency response metrics:

- **Passive System:**

- Peak magnitude: 5.98 dB at 1.21 Hz
- Natural frequency clearly visible in magnitude plot
- Limited damping at resonance
- Poor high-frequency attenuation

- **Active System:**

- Peak magnitude: −9.26 dB at 1.34 Hz
- Significant attenuation at resonant frequency
- Improved damping characteristics
- Better high-frequency isolation

- **Improvement:**

- Peak reduction: 15.24 dB
- Vibration amplitude reduction: approximately 82%
- Enhanced performance across critical frequency range (0.5-10 Hz)

4.4 Performance Metrics

Table 4 summarizes the quantitative performance comparison between passive and active suspension systems.

Table 4: Comprehensive performance metrics comparison

Metric	Passive	Active
Frequency Domain		
Peak magnitude (dB)	5.98	−9.26
Peak frequency (Hz)	1.21	1.34
Magnitude improvement (dB)	–	15.24
Amplitude reduction (%)	–	82.0
Time Domain		
Peak body acceleration (m/s ²)	12.0	6.0
Acceleration reduction (%)	–	50.0
Settling time (s)	4.0	1.5
Settling time reduction (%)	–	62.5
Maximum suspension travel (m)	0.08	0.05
Travel reduction (%)	–	37.5

5 Results and Discussion

5.1 Frequency Response Analysis

The Bode plot analysis reveals several important characteristics:

1. **Resonance Suppression:** The active system successfully suppresses the resonance peak, reducing the magnitude by 15.24 dB. This translates to significantly reduced body oscillations when the vehicle encounters periodic disturbances at the natural frequency.
2. **Phase Behavior:** The active controller modifies the phase characteristics, improving the system's ability to anticipate and counteract disturbances.
3. **Stability Margins:** Both systems exhibit stable behavior with adequate gain and phase margins.

5.2 Time Domain Performance

The time-domain simulations provide clear evidence of improved performance:

1. **Body Acceleration:** The active system reduces peak body acceleration by 50%, directly improving passenger comfort.
2. **Settling Time:** A 62.5% reduction in settling time means the vehicle returns to steady state much faster after encountering road disturbances.
3. **Suspension Travel:** The 37.5% reduction in suspension travel indicates better utilization of available suspension stroke while maintaining ride quality.

5.3 Physical Interpretation

The performance improvements can be explained as follows:

- The PID controller continuously monitors body velocity
- When upward velocity is detected, the actuator generates downward force
- This creates effective damping without the energy dissipation losses of passive dampers
- The integral action ensures zero steady-state error for sustained disturbances
- Derivative action provides predictive damping, anticipating future motion

5.4 Practical Considerations

Advantages of active suspension:

- Superior ride comfort through vibration isolation (82% amplitude reduction)
- Improved handling and vehicle stability
- Faster disturbance rejection (62.5% faster settling)
- Reduced body acceleration for passenger comfort (50% reduction)
- Adaptability to different road conditions

- Independent optimization of comfort and handling

Challenges and limitations:

- Increased system complexity requiring sophisticated control algorithms
- Power consumption requirements for actuators
- Actuator bandwidth and force limitations (± 5000 N in this design)
- Higher manufacturing and maintenance costs
- Reliability considerations for electronic components
- Sensor requirements for velocity feedback

5.5 Design Trade-offs

The controller design involves balancing multiple objectives:

1. **Ride Comfort vs. Handling:** Higher damping improves comfort but may reduce tire contact and handling. The PID controller achieves a good balance.
2. **Performance vs. Actuator Power:** Aggressive control (higher gains) requires larger actuator forces. The selected gains provide good performance within ± 5000 N limits.
3. **Complexity vs. Robustness:** The PID controller offers a good compromise between performance and implementation complexity.

6 Conclusions

6.1 Summary of Findings

This study successfully demonstrates the design and analysis of an active suspension system using PID control. The key findings are:

1. The quarter-car mathematical model accurately represents suspension dynamics for analysis and design purposes.
2. The designed PID controller with gains $K_p = 5000$, $K_i = 100$, and $K_d = 2000$ achieves:
 - 15.24 dB reduction in peak frequency response magnitude
 - 82% reduction in vibration amplitude
 - 50% reduction in peak body acceleration
 - 62.5% reduction in settling time
3. Frequency domain analysis confirms improved performance across the critical frequency range of 0.5-10 Hz, covering typical road disturbances.

4. Time domain simulations demonstrate superior transient response and disturbance rejection compared to passive suspension.
5. The active system maintains stability while significantly enhancing ride comfort and vehicle dynamics.

6.2 Contributions

This project contributes to:

- Practical understanding of active suspension control principles
- Hands-on experience with PID control for mechanical systems
- Development of MATLAB/Simulink modeling skills
- Comparative analysis methodology for evaluating suspension performance
- Foundation for exploring advanced control strategies

6.3 Learning Outcomes

Through this project, the following skills were developed:

- Mathematical modeling of complex mechanical systems
- Controller design and tuning techniques
- Simulation and analysis using MATLAB/Simulink
- Frequency and time domain analysis methods
- Technical documentation and report writing
- Interpretation of simulation results

6.4 Future Work

Potential extensions of this work include:

1. Advanced Control Strategies:

- LQR (Linear Quadratic Regulator) for optimal performance
- Model Predictive Control (MPC) with constraint handling
- Adaptive control for parameter variations
- Robust H-infinity control for uncertainty
- Fuzzy logic or neural network controllers

2. Full Vehicle Model:

- Extension to full-car (7-DOF) model
- Pitch and roll dynamics inclusion

- Inter-wheel coordination strategies
- Load transfer analysis

3. Real-World Validation:

- Hardware-in-the-loop (HIL) testing
- Experimental prototype development
- Road testing with real vehicles
- Validation with measured road profiles

4. Semi-Active Solutions:

- Magnetorheological (MR) dampers
- Variable damping strategies
- Reduced power consumption
- Cost-effective implementation

5. Advanced Features:

- Preview control using road sensors
- Sky-hook damping strategies
- Multi-objective optimization
- Adaptive gain scheduling

6.5 Final Remarks

The active suspension system designed in this project demonstrates the significant advantages of feedback control in automotive applications. The quantitative results validate theoretical predictions and confirm the feasibility of active suspension technology for improving vehicle performance.

The 82% reduction in vibration amplitude and 62.5% faster settling time represent substantial improvements that would translate to noticeably better ride comfort and handling in real-world driving conditions. The methodologies developed here provide a solid foundation for further research and practical implementation of advanced suspension systems in modern vehicles.

The modular Simulink implementation allows for easy modification and extension, making it an excellent platform for exploring additional control strategies and system enhancements. This project successfully demonstrates the practical application of control theory principles learned in the Systems Simulation Laboratory course.

Acknowledgments

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Special thanks to the Systems Simulation Laboratory staff for their assistance with MATLAB/Simulink implementation and troubleshooting.

A MATLAB Code

A.1 System Parameters

```

1 %% Suspension System Parameters
2 % Quarter-car model physical parameters for active suspension design
3
4 % Masses (kg)
5 ms = 300;      % Sprung mass - vehicle body
6 mus = 40;      % Unsprung mass - wheel assembly
7
8 % Suspension parameters
9 ks = 18000;    % Suspension spring stiffness (N/m)
10 cs = 1200;    % Suspension damping coefficient (N*s/m)
11
12 % Tire parameters
13 kt = 195000;   % Tire stiffness (N/m)
14 ct = 75;      % Tire damping coefficient (N*s/m)
15
16 % Controller gains (PID)
17 Kp = 5000;     % Proportional gain
18 Ki = 100;     % Integral gain
19 Kd = 2000;     % Derivative gain
20
21 % Simulation parameters
22 Ts = 0.001;    % Sample time for discrete blocks (1 ms)

```

Listing 1: suspension_parameters.m - System parameter definitions

A.2 Bode Plot Analysis Code

```

1 %% Bode Plot Comparison: Passive vs Active
2 clc; clear; close all;
3
4 % Load system parameters
5 suspension_parameters;
6
7 %% Build Transfer Functions
8 s = tf('s');
9
10 % Passive System Transfer Function
11 %  $H(s) = (cs*s + ks)(ct*s + kt) / [ms*mus*s^4 + \dots]$ 
12 num_passive = conv([cs, ks], [ct, kt]);
13 den_passive = [ms*mus, (ms*ct + mus*cs), ...
14               (ms*kt + mus*ks + cs*ct), ...
15               (cs*kt + ct*ks), ks*kt];
16 H_passive = tf(num_passive, den_passive);
17
18 % PID Controller with derivative filter
19 N = 100; % Filter coefficient to prevent noise amplification
20 C_PID = Kp + Ki/s + (Kd*N*s)/(s + N);
21
22 % Active System - Closed loop with velocity feedback
23 H_velocity = s; % Velocity = s * Position
24 H_active = feedback(H_passive, C_PID * H_velocity);

```

```

25
26 %% Generate Bode Plot
27 figure('Position', [100, 100, 1200, 800]);
28
29 % Frequency range (logarithmic spacing)
30 w = logspace(-2, 3, 1000); % 0.01 to 1000 rad/s
31
32 % Get frequency responses
33 [mag_p, phase_p] = bode(H_passive, w);
34 [mag_a, phase_a] = bode(H_active, w);
35
36 % Squeeze dimensions for plotting
37 mag_p = squeeze(mag_p);
38 phase_p = squeeze(phase_p);
39 mag_a = squeeze(mag_a);
40 phase_a = squeeze(phase_a);
41
42 % Magnitude plot
43 subplot(2,1,1);
44 semilogx(w, 20*log10(mag_p), 'b-', 'LineWidth', 2.5);
45 hold on;
46 semilogx(w, 20*log10(mag_a), 'r-', 'LineWidth', 2.5);
47 grid on;
48 ylabel('Magnitude (dB)', 'FontSize', 13);
49 title('Bode Diagram - Magnitude', ...
50       'FontSize', 14, 'FontWeight', 'bold');
51 legend('Passive System', 'Active System', ...
52       'Location', 'best', 'FontSize', 12);
53 xlim([w(1) w(end)]);
54
55 % Phase plot
56 subplot(2,1,2);
57 semilogx(w, phase_p, 'b-', 'LineWidth', 2.5);
58 hold on;
59 semilogx(w, phase_a, 'r-', 'LineWidth', 2.5);
60 grid on;
61 xlabel('Frequency (rad/s)', 'FontSize', 13);
62 ylabel('Phase (degrees)', 'FontSize', 13);
63 title('Bode Diagram - Phase', ...
64       'FontSize', 14, 'FontWeight', 'bold');
65 legend('Passive System', 'Active System', ...
66       'Location', 'best', 'FontSize', 12);
67 xlim([w(1) w(end)]);
68
69 sgtitle(['Passive vs Active Suspension System -', ...
70         ' Bode Plot Comparison'], ...
71         'FontSize', 16, 'FontWeight', 'bold');
72
73 %% Calculate and Display Performance Metrics
74 [peak_p, idx_p] = max(mag_p);
75 [peak_a, idx_a] = max(mag_a);
76
77 fprintf('\n===== \n');
78 fprintf('    BODE PLOT COMPARISON RESULTS \n');
79 fprintf('===== \n \n');
80
81 fprintf('PASSIVE SYSTEM: \n');
82 fprintf(' Peak Magnitude: %.2f dB at %.2f Hz \n', ...

```

```

83         20*log10(peak_p), w(idx_p)/(2*pi));
84 fprintf(' Peak Frequency: %.2f rad/s\n', w(idx_p));
85
86 fprintf('\nACTIVE SYSTEM:\n');
87 fprintf(' Peak Magnitude: %.2f dB at %.2f Hz\n', ...
88         20*log10(peak_a), w(idx_a)/(2*pi));
89 fprintf(' Peak Frequency: %.2f rad/s\n', w(idx_a));
90
91 fprintf('\nPERFORMANCE IMPROVEMENT:\n');
92 fprintf(' Peak Reduction: %.2f dB\n', ...
93         20*log10(peak_p) - 20*log10(peak_a));
94 fprintf(' Amplitude Reduction: %.1f%%\n', ...
95         (1 - peak_a/peak_p)*100);
96
97 fprintf('\n===== \n');
98
99 % Save figure
100 saveas(gcf, 'bode_comparison.pdf');
101 saveas(gcf, 'bode_comparison.png');

```

Listing 2: Bode plot comparison and performance metrics calculation

B Simulink Model Details

B.1 Subsystem Descriptions

Road Profile Subsystem: Generates three types of road inputs selectable via Manual Switch:

- Step bump: 5 cm amplitude at $t = 1$ s
- Sinusoidal: 3 cm amplitude, 1 Hz frequency
- Random roughness: Band-limited white noise

Tire Dynamics Subsystem: Implements the tire force equation:

$$F_{tire} = k_t(x_{us} - x_r) + c_t(\dot{x}_{us} - \dot{x}_r)$$

Suspension Dynamics Subsystem: Calculates suspension force:

$$F_{susp} = k_s(x_s - x_{us}) + c_s(\dot{x}_s - \dot{x}_{us})$$

Mass Dynamics Subsystems: Integrate forces using Newton's second law with double integration to obtain position and velocity.

PID Controller Subsystem: Implements velocity feedback control with saturation (± 5000 N) and rate limiting (± 50000 N/s).

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